

Element B: Documentation and Analysis of Prior Design Attempts

Prior Solution #1 - Wildlife Toybox Plastic Tub + Heavy Objects



Details: This toy has previously been used for enrichment of the leopards in the Nashville Zoo by putting food under it and stacking heavy objects like rocks and tree logs on top of it to hold it down. The tub is made out of a thick plastic material that is safe for the animals and is durable enough to last a very long time. While we couldn't find where to buy the toy itself, the Zookeeper told us that they bought it off of Wildlife Toybox which leads us to conclude that the website no longer sells it.

Analysis: This is a simple and solid puzzle for the leopards that is easy to set up/clean; however, as described by our zookeeper stakeholder, the time it takes for them to solve it varies wildly depending on what is stacked on top of it, in one instance even leading to Jabari giving up after working on the puzzle over night (also said that Jabari made it harder on himself by flipping a

water bowl over the tub). Also since the Zoo already owns one of these tubs as a toy, it would be beneficial to design ours to stimulate a different problem solving behavior in the leopards.

Prior Solution #2 - Wildlife Toybox Plastic Cat Toy



Design: This is another toy offered by Wildlife Toybox specifically marketed for large cats. The toy is made out of a thick plastic material that is safe for the animals and is durable enough to last a very long time. You can also pay extra to include a scent or a jingle element in the toy. The toy retails for about 330\$ on the website www.wildlifetoybox.com under the name Cat Toy.

Analysis: While this is also a very high quality toy specifically designed for large cats, even its cheapest option is above our budget of 250\$. Also, considering the method that the zookeepers use to clean out toys using a hose, this toy seems more difficult to clean than most.

Prior Solution #3 - Big Cat Sanctuary Homemade Enrichment Toy



Design: This is a homemade toy created by The Big Cat Sanctuary in Kent, UK using a wooden pole and broom heads. It is shown being used by a white tiger named Laila, who bats at the wooden pole hanging on a chain.

Analysis: This is a very cost effective design that uses easily available materials. It also might be possible to have the broom heads emit a scent when scratched, but it would probably not act well as an actual feeder. Not only would this toy be difficult to clean (chunks in between the bristles), it also lacks any sort of puzzle or problem solving element. Additionally, the broom stick heads and wood might splinter or carry bacteria if not properly treated and disinfected, posing a potential safety threat to the leopards and ruining the longevity of the toy.

Prior Solution #4 - Amazon Dog Feeder Puzzle



Design: This is a toy sold on Amazon by DR CATCH as a puzzle feeder for small dogs. While not explicitly meant for cats, it serves many of the functions that our design must fulfill, for example being solvable repeatedly without becoming too easy. The toy features sliding plastic doors that can cover compartments with food in them, forcing the animal to cover some compartments to get food from others. The toy retails for about 10\$ on www.amazon.com.

Analysis: This toy can provide good inspiration for a more puzzle-like design with several moving parts, however such a design would be more prone to breaking and more difficult to create without manufacturing the parts. In addition to these points, such a design would be difficult to clean as food and fluids could get stuck in the sliding components. In order to make this design more viable, it would have to be scaled up and implemented in a way that better conforms to the constraints of the project.

Prior Solution #5 - Suspended Barrel Roller Enrichment Toy



Design: This concept is inspired by observations of how big cats like tiger interact with objects during play, as described by Michael Broad in his article *How Do Tigers Play?*. The image shows a large cat engaging with a cylindrical roller, which aligns with documented behaviors such as pushing, clawing, biting, and balancing on moving objects. The proposed design consists of a heavy-duty cylindrical barrel made from thick, non-toxic plastic or reinforced rubber, mounted horizontally on a fixed axle or partially suspended system. The barrel can rotate when pushed or climbed on. Small holes can be added to dispense food or scent intermittently as the animal manipulates the toy. The structure would be anchored securely to prevent tipping or detachment but still allow smooth rotational movement.

Analysis: This design effectively mimics natural play behaviors described in the source, encouraging physical activity (climbing, balancing) and cognitive engagement (figuring out how to release food). Unlike static toys, the rolling motion introduces unpredictability, which can sustain interest over longer periods. Compared to Prior Solution #1, this toy introduces dynamic movement rather than resistance-based problem solving. Compared to #3, it is significantly more

durable and hygienic. While slightly more complex than #4, it avoids fragile small parts by relying on large-scale movement instead of intricate mechanisms. Potential drawbacks include the need for secure installation and ensuring the barrel cannot be dislodged. However, overall, this design provides a balanced combination of durability, enrichment variety, and ease of maintenance, making it a strong candidate within the project constraints.

Citations

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